



**Kyle Stang**  
**Final Intern Presentation**

August 18<sup>th</sup>, 2026

## Background Information

- **Dual Bachelor of Science:**
  - School: UMass Dartmouth
  - Majors: Electrical & Computer Engineering
  - Expected Graduation Date: May 2027
- Work Facility: Taunton, MA
- Manager/Supervisor: David Barboza
- Mentor(s): David Barboza, Richard Morani, Bob Nadeau, Matt King, Alex Wengryn
- Department: Hammerhead HW & Systems Engineering



# Tasking

- Aid Hammerhead's EE related development
- Work alongside and learn from the larger Hammerhead HW/Systems team
- Gain practical experience in systems-level engineering and design

## Measures of Success

- Number of new product requests entered into the Windchill database
- Drawings revised and created
- Potential resources saved in different test procedures

# Results & Deliverables

## Task 1 – Aiding EE Development

- Designed and created drawings for a new test cable, optimizing and saving costs in Hammerhead FAT procedures
  - Detailed Visio cable drawing
  - BOM
  - Circuit Diagram & Splice Table
- Created 4 NPRs for Hammerhead PCB development, including 300+ new components added to the Windchill database
- Designed concept for an expanded BCD (Battery Charge Device) including an added 4<sup>th</sup> battery charging section
- Redesigned Hammerhead state machine with improved readability and detail
- Tested and analyzed cables and cable designs to diagnose and fix test errors
- Performed regular shortage and continuity tests

# Results & Deliverables

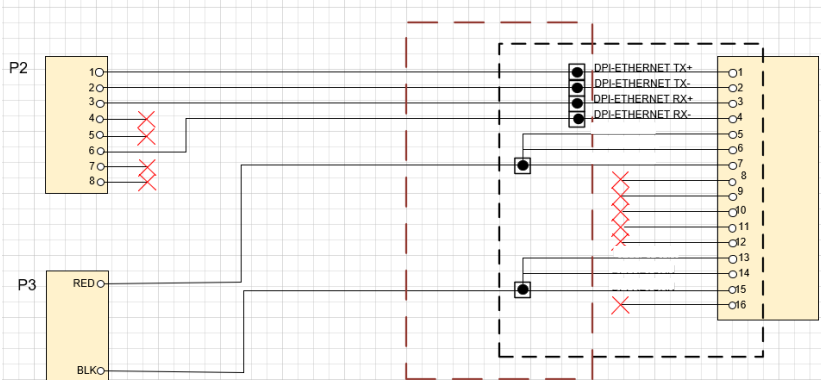
## **Task 2 – Working Alongside the Hammerhead Team**

- Worked within Hammerhead's I&T and T&E teams
- Collaborated on multiple different projects being done in the office, the Systems Integration Lab (SIL), and the Hammerhead operations floor

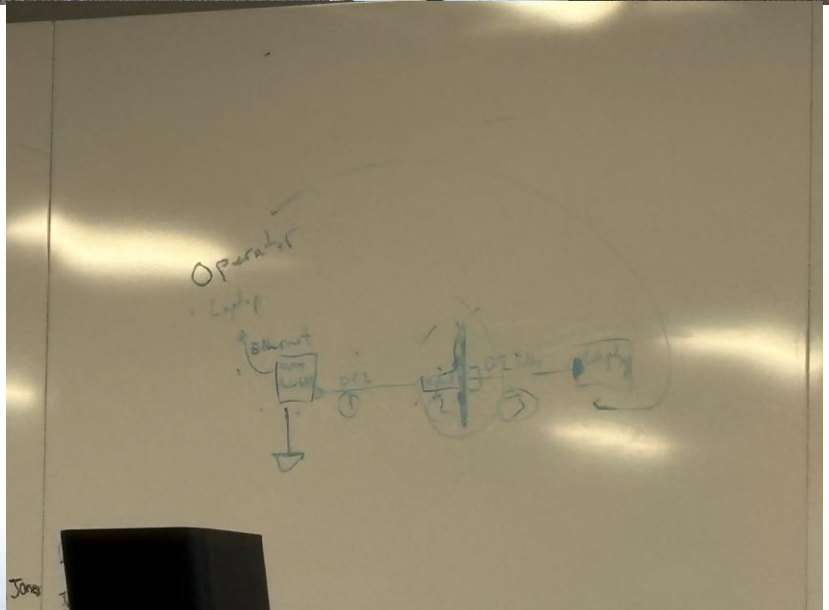
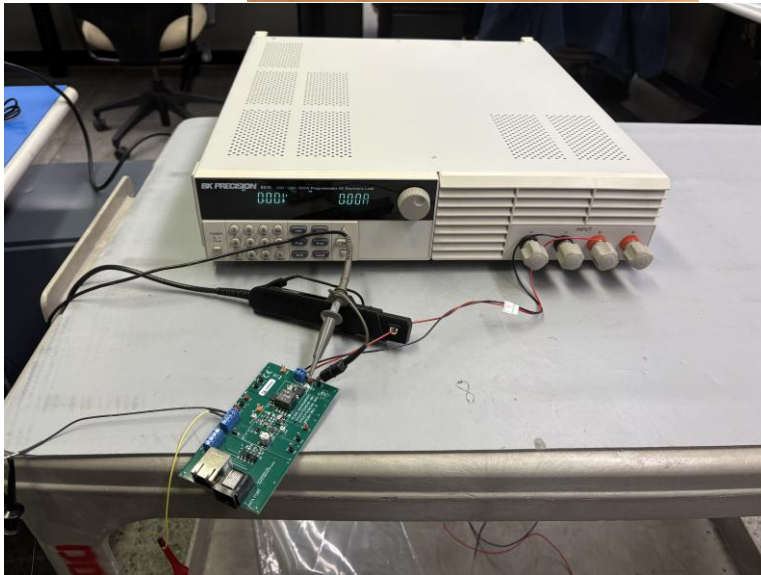
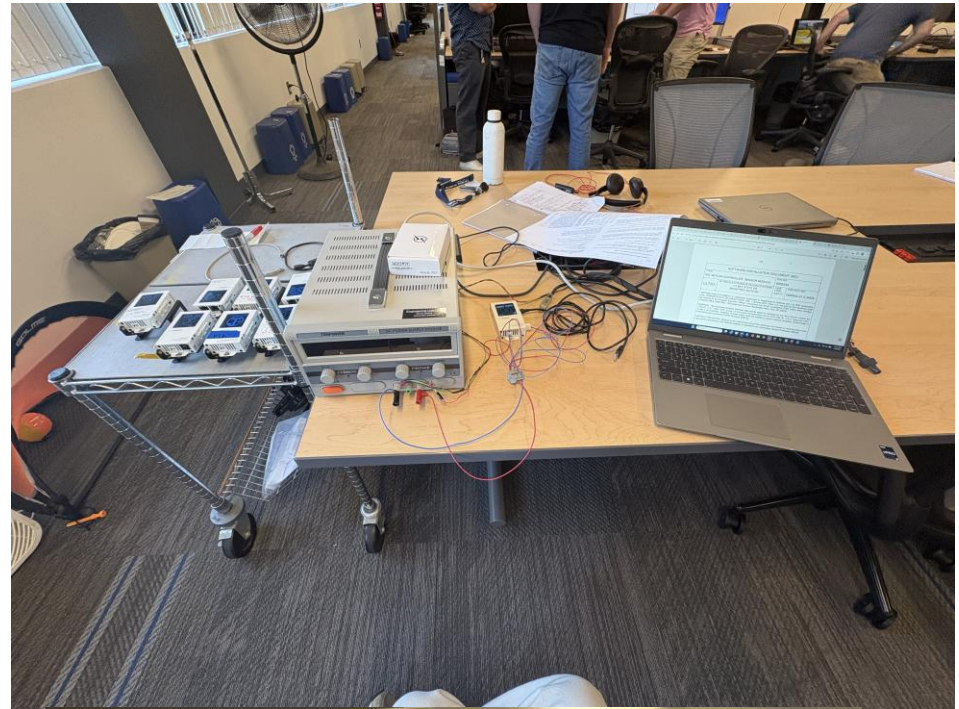
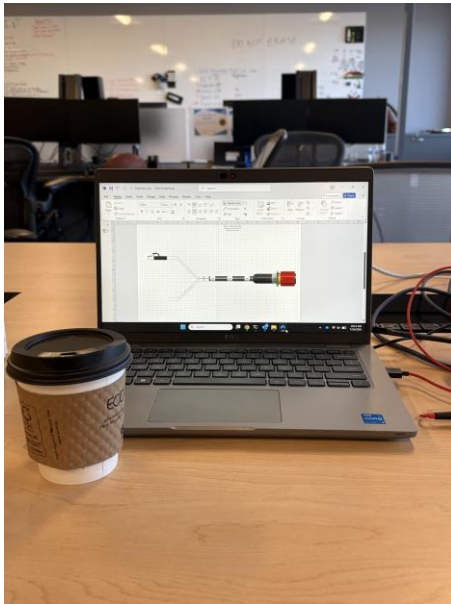
## **Task 3 – Gaining Systems Experience**

- Was required to learn the entire Hammerhead system
  - State machine redesign
  - Cable error analysis and cable redesign within the larger system

**GENERAL DYNAMICS**  
Mission Systems



# Pictures



**GENERAL DYNAMICS**  
Mission Systems



# Community Engagement

- Intern Program Events
  - Enterprise-Wide Intern Events (Meeting former interns, learning about ELP/MLP, Mapping my career journey, Resume and Interview tips)
- Other
  - Tri-Weekly Standup Meetings for Hammerhead
  - All Hands Meetings
  - Group Lunches
  - Tours of SIL and Ops floor
  - Sharing news on LinkedIn



# My Internship Expectations

- A new environment to provide a new look at industry work
- Learn more about defense contracting
- An opportunity to apply my school knowledge
- A transition from government funded programs to private sector work

## What I Did To Meet My Expectations

- What did you do to meet both your expectations, and the expectations your manager set for you?
  - Ask for newer and more challenging projects to work on
  - Diversified the type of work I was used to with the Systems team
- What advice were you provided that you found helpful?
  - Walk around, ask for help & watch what other people are doing